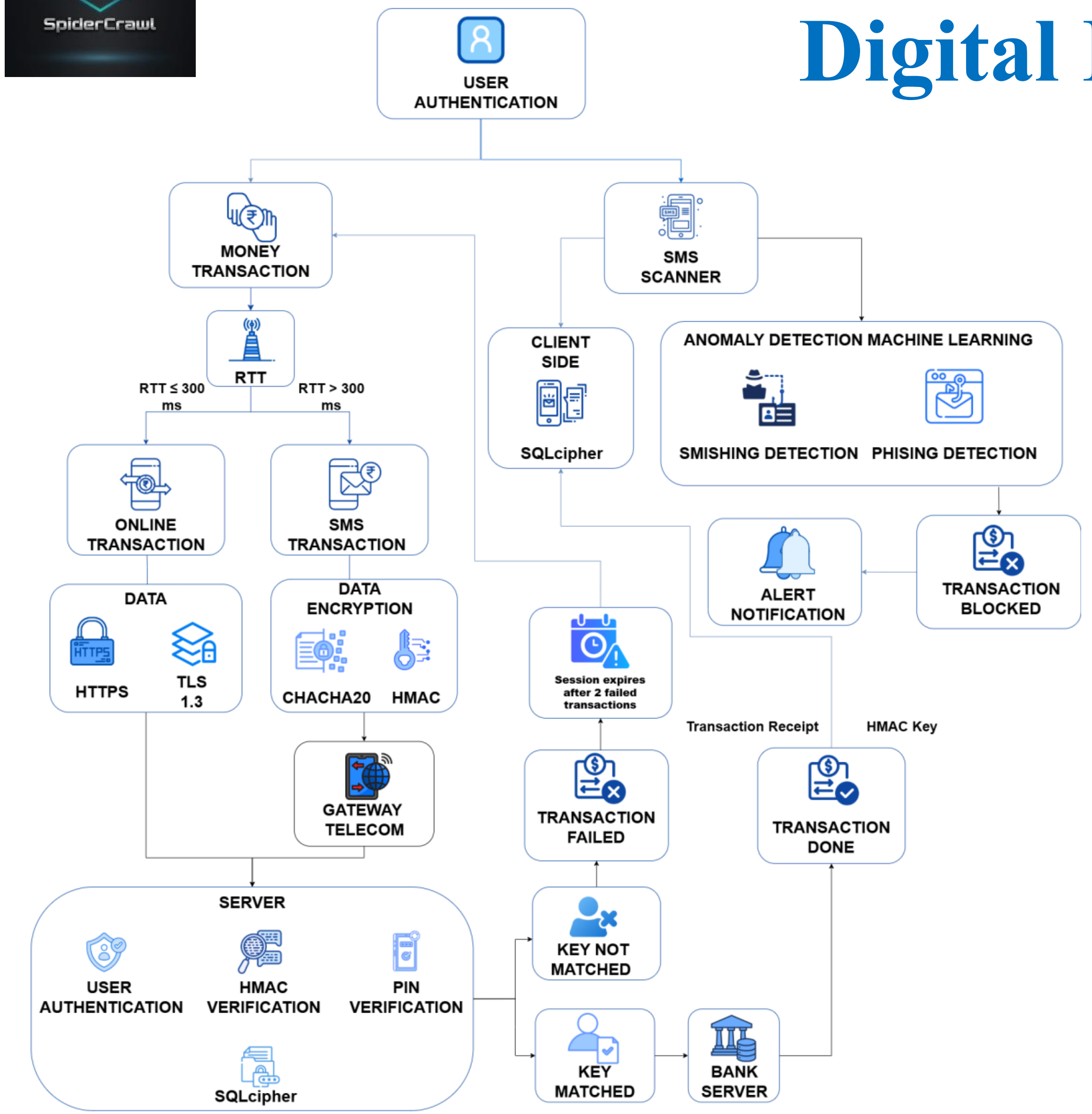
SMART INDIA HACKATHON 2025

TITLE PAGE

- **Problem Statement ID – 25205**
- **Problem Statement Title- Cybersecurity Framework for Rural Digital Banking**
- **Theme- Blockchain & Cybersecurity**
- **PS Category- Software**
- **Team ID- K25053**
- **Team Name- SpiderCrawl**

Cybersecurity Framework for Rural Digital Banking



PROBLEM RESOLUTION

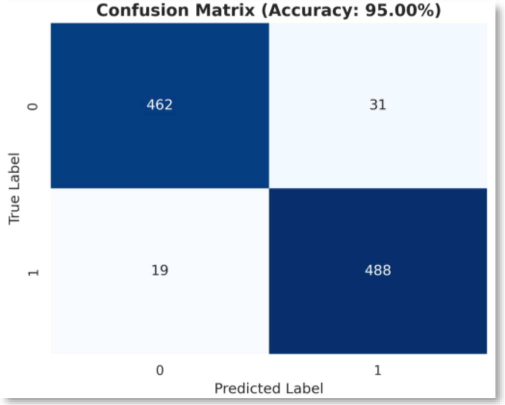
Existing tools/Platform	Limitations/Issues	How our solution helps
PhonePe	TLS-only encryption; slower in low-network conditions	Hybrid TLS + ChaCha20 encryption
Google Pay	Single-layer verification; OTP can be intercepted	Multi-layer verification (PIN + HMAC + device binding)
Paytm	Focused only on online transactions; no fallback In low-network areas	Offline USSD + hybrid ChaCha20 layer enables secure access even in low network connectivity

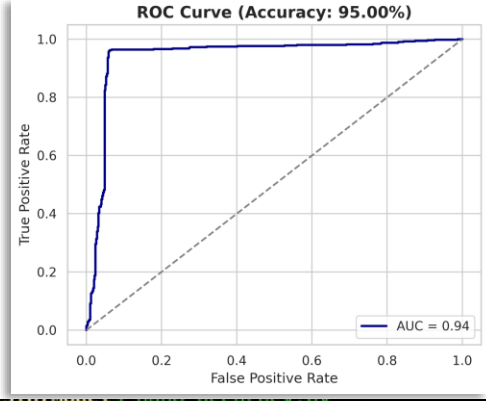
UNIQUE VALUE PROPOSITIONS (UVPs)

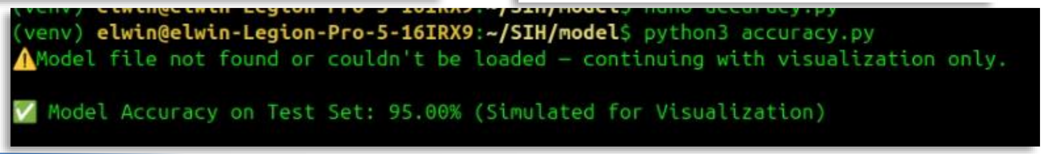
- **Uninterrupted Banking:** Seamless hybrid transactions via **RTT** for any network.
- **Robust Data Protection:** End-to-end **USSD chachas20-poly1305, HMAC & SQLCipher** encryption.
- **TLS 1.3** & **HMAC Dual-Protection** — ensures **Secure Handoff** and **Layered Online Security** between client and server.

TECHNICAL APPROACH

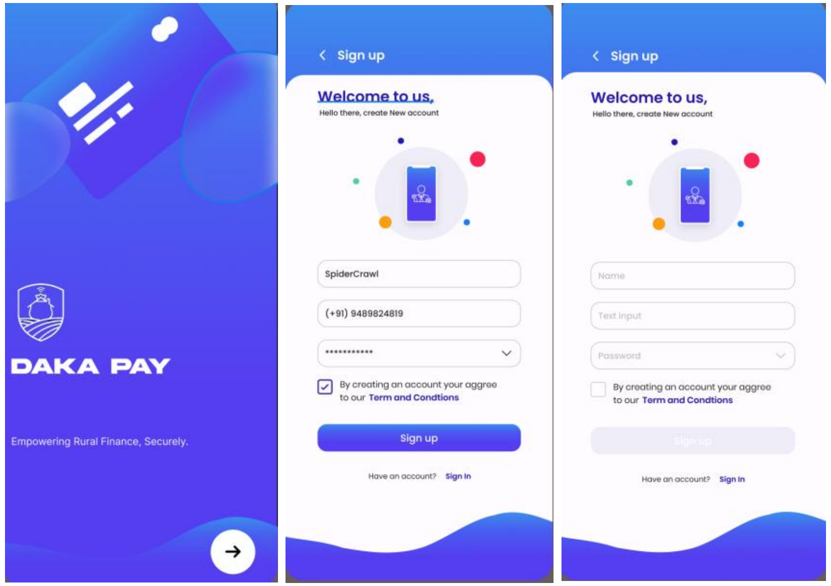
Current Status: 57% of android application is completed with fully trained DL. Yet to release the interface on regional language







SCREENSHOTS



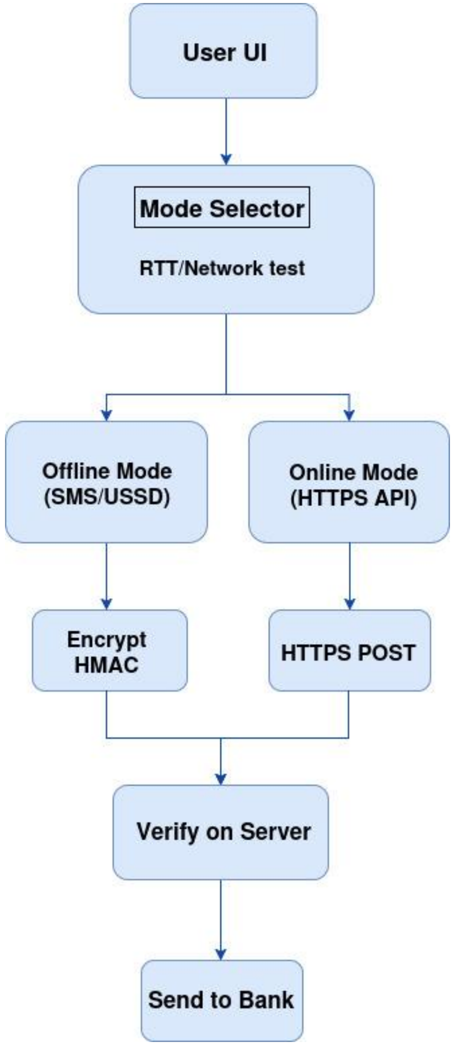
TECHNOLOGY USED

Technolgy	Website/Softwares/Language
Frontend	React
Backend	Kotlin, JAVA,Flutter
ML/DL	Python, CNN

LINKS

CLICK BELOW

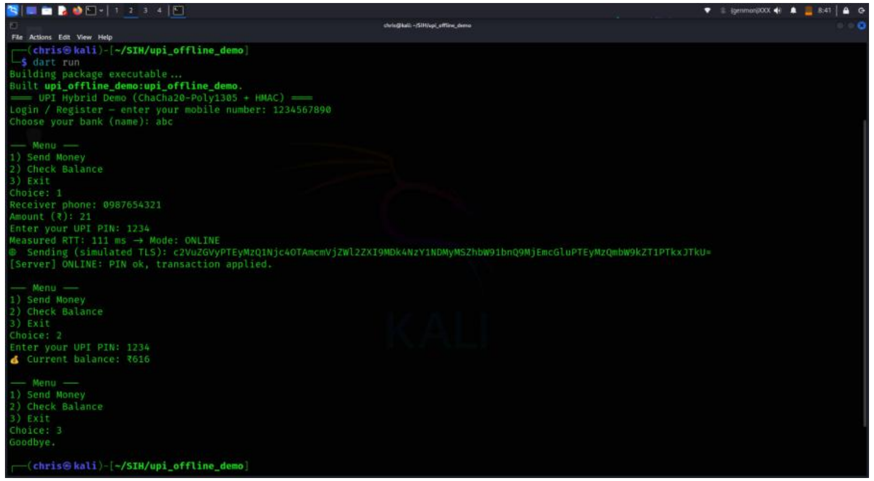
FLOWCHART



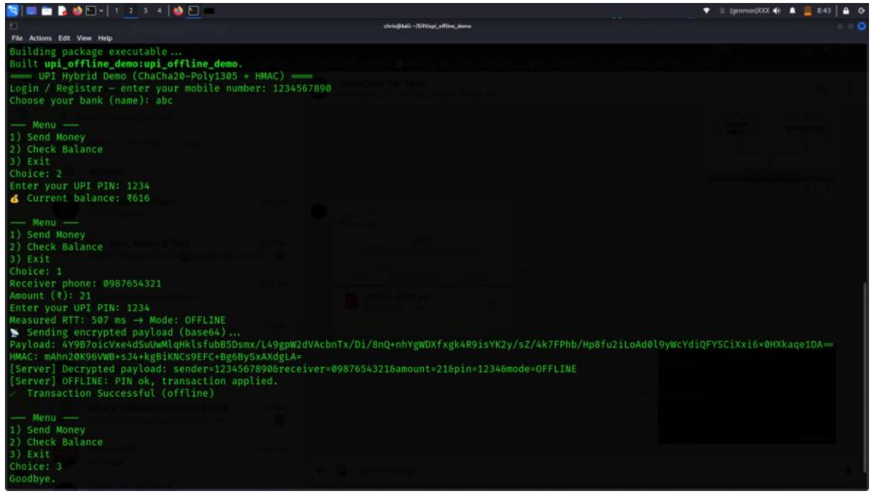
FEASIBILITY AND VIABILITY

Aspect	Technical	Analysis of the Feasibility of the Ide	Potential Challenges & Risks	Strategies for Overcoming These Challenge
Feasibility	TLS 1.3 encryption for online transactions; ChaCha20 + HMAC for offline USSD transactions; SQLCipher for local data protection	System works in both high and low network conditions. Simple flow is suitable for rural and low-connectivity users.	Synchronizing online and offline transaction modes; Ensuring secure key handling and verification	Strengthen hybrid encryption flow (TLS + ChaCha20 + SQLCipher); Implement proper key verification
Risks and Challenges	DL model for phishing/smishing detection; Lightweight encryption ensures smooth use on low-end devices	DL improves fraud detection if trained well	DL accuracy & false positives; Phishing/smishing threats; Performance issues on low-end devices; Data privacy compliance	Regular DL model training; Optimize for lightweight performance; Follow data security and privacy complianc
Operational Strategies to overcome	HMAC verification ensures transaction integrity; DL anomaly detection alerts suspicious activity;	Supports online and offline transactions effectively. Automated retries handle most processes	Session expiry is handled automatically	The flow handles key verification, session management (including automatic handling of session expiry), HMAC validation, and DL-based anomaly detection automatically.

ONLINE TRANSACTION



OFFLINE TRANSACTION



IMPACT AND BENEFITS

POSITIVE IMPACT

- **Financial Inclusion:** Expands banking access to **rural underserved**.
- **Economic Empowerment:** Boosts **local commerce** & productivity via easy transactions.
- **Enhanced Security: Fraud detection** (Smishing/Phishing) & **encryption** build trust.
- **Outage Resilience:** Ensures **transactions even during outages** (via offline ops).
- **Digital Literacy Boost:** Encourages **tech adoption** and financial awareness.
- **Transparency & Trust:** Clear digital records foster **accountability** and trust.
- **Accessibility for All:** Supports **diverse users** regardless of internet literacy.

BENEFITS

Social	Economic	Environmental
Ensures secure user authentication	Enables secure transaction verification	Minimizes paper usage
Provides notifications to users	Speeds up transaction processing	Reduces travel for offline transactions
Provides access through multiple digital channels	Supports both online and offline transaction handling	Optimizes system resource use via automated DL/ML checks

NEGATIVE IMPACT

- **Digital Divide:** Challenges with **digital literacy** and **smartphone dependency**.
- **Security Vigilance:** Ongoing need for **data privacy** and robust **security updates**.

FUTURE SCOPE

- **Govt. Scheme Integration:** Link with **rural welfare programs**.
- **Digital Wallet:** Expand to **QR-based merchant payments**.
- **Financial Literacy:** In-app **educational modules**.
- **Biometric Auth:** Enhance security with **fingerprint/face ID**.

RESEARCH AND REFERENCES

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Research Paper Links	Brief Description
Designing Resilient Systems for Low-Connectivity Environments: A Comparative Study of Progressive Web Applications and Distributed Systems (Google Scholar) https://www.diva-portal.org/smash/record.jsf?dswid=9531&pid=diva2%3A1959298	This paper likely explores how software systems can be designed to function reliably in environments with poor or unstable network connectivity.
Holistic Review of Modern Cryptographic Algorithms and Their Application in Securing Mobile Apps, APIs, and Databases (ResearchGate)	This paper reviews cryptographic methods for mobile apps, APIs, and databases, supporting secure authentication and data protection.
AI Driven Fraud Detection Models in Financial Networks: A Comprehensive Systematic Review (IEEE) https://ieeexplore.ieee.org/abstract/document/11113282	This paper reviews AI and DL techniques for real-time fraud detection, supporting the anomaly detection step in your transaction flow.
Offline Transaction System https://doi.org/10.1051/itmconf/20224403072	This paper focuses on enabling secure and reliable offline payment systems to expand digital transactions in low-network and rural areas.
A Multi-Channel System Architecture for Banking https://papers.ssrn.com/sol3/papers.cfm?abstract_id=3571389	This paper proposes a multi-channel architecture for banking, guiding the integration of new technologies while maintaining core financial services across various access channels.